Indian Statistical Institute, Bangalore
B. Math.
First Year, First Semester

Analysis I — Instructor: B V Rajarama Bhat
Mid-term examination — Date : Sept. 8, 2025
Total Marks: 105 Maximum marks: 100 — Time: 3 hours

(1) Let $\mathcal{P}(A)$ denote the power set of a non-empty set A. Let $f : A \to \mathcal{P}(A)$ be a function. (i) Show that f is not surjective. (ii) Show that $f^{-1}(B)$ is empty for some non-empty subset B of $\mathcal{P}(A)$. [15]

(2) (i) Write down a bijection between $(0, 1]$ and $[0, 1]$. (ii) Write down a bijection between $[0, 1]$ and $[0, \infty)$. [15]

(3) Show that given any real number x there exists a natural number n such that $n > x$.
[15]

(4) Let $\{a_n\}_{n\in\mathbb{N}}, \{b_n\}_{n\in\mathbb{N}}$ be two sequences of real numbers. Define a new sequence $\{c_n\}_{n\in\mathbb{N}}$ by

$$c_n = \begin{cases} a_n & \text{if } n \text{ is odd;} \\ b_n & \text{if } n \text{ is even.} \end{cases}$$

Suppose $\{a_n\}_{n\in\mathbb{N}}$ converges to x and $\{b_n\}_{n\in\mathbb{N}}$ converges to y. Show that $\{c_n\}_{n\in\mathbb{N}}$ is convergent if and only if $x = y$. [15]

(5) Let A be a non-empty subset of $\mathbb{R}$ and let $f : A \to \mathbb{R}$ be a function. Define $g : A \to \mathbb{R}$ by $g(x) = |f(x)|,\ \forall x \in A$. (i) Show that if f is continuous then g is continuous. (ii) Show that it is possible that g is continuous but f is not continuous. [15]

(6) Define Cauchy property for sequences of real numbers. Show that every Cauchy sequence of real numbers is bounded. [15]

(7) Let $a, b \in \mathbb{R}$ with $a < b$. Let $f : [a, b] \to \mathbb{R}$ be an increasing function. Fix $c \in (a, b)$. Take

$$A = \{f(x) : x \in [a, c)\},\ B = \{f(y) : y \in (c, b]\}.$$

(i) If f is continuous at c show that

$$f(c) = \sup(A) = \inf(B).$$

(ii) If

$$\sup(A) = \inf(B),$$

show that f is continuous at c. [15]